Register Number:

Name of the Candidate:

0318-A

B.E. DEGREE EXAMINATION, 2021

(COMPUTER SCIENCE AND ENGINEERING)

(FOURTH SEMESTER)

CSBS-401/ITBS-401/AIBS-401: DISCRETE MATHEMATICS

(Common with Information Technology and Artificial Intelligence and ML)

(With effect from May 2021)

May] [Time : 3 Hours

Maximum : 75 Marks

SECTION – A

Answer ALL questions **(10 x 1=10)**

1.	*What is mean by tautology?*	*CO1*	*K1*
2.	*Define Rule P and Rule CP in the inference theory*	*CO1*	*K1*
3.	*If A={1} and B={1,2} find A-B.*	*CO2*	*K1*
4.	*Define disjoint sets with a example.*	*CO2*	*K1*
5.	*Define a Lattice.*	*CO3*	*K1*
6.	*State the isotonic property of a Lattice.*	*CO3*	*K1*
7.	*Define a semi group with an example.*	*CO4*	*K1*
8.	*what is the identity element of the algebraic system* $\{Q^+,*\}$ *Where* Q^+ *is the set of positive rational numbers and * is defined by* $a * b = \frac{ab}{2}$ *for all* $a,b \in Q^+$.	*CO4*	*K1*
9.	*Define complete graph with an example.*	*CO5*	*K1*
10.	*Define spanning subgraph.*	*CO5*	*K1*

SECTION – B

Answer either (a) or (b) in each Question **(5 x 3=15)**

11 (a) *Prove that* $\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q)) \Leftrightarrow (\neg P \vee Q)$ *CO1* *K2*

OR

(b) *Prove that* $(P \rightarrow Q) \wedge (Q \rightarrow R) \Rightarrow (P \rightarrow R)$ *CO1* *K2*

12*(a) If A,B and C are any three sets Prove that* $(A \cap B) - C = (A - C) \cap (B - C)$ *CO2* *K2*

OR

(b) If R and S be relations on a set A represented by the matrices $M_R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ *and* $M_s = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

Find $R \circ S$, $S \circ R$ *and* $R \oplus S$ *CO2* *K2*

13*(a) Draw the Hasse diagram representing the partial ordering* $\subseteq$ *on the set P(S), where* $S = \{1,2,3\}$ *CO3* *K2*

OR

(b) State and prove the Modular inequality of a Lattice *CO3* *K2*

14*(a) Prove that* $\{Z_6, \times_6\}$ *is a monoid.* CO4 K2

OR

(b) Show that $\{(1,-1,i,-i),\times\}$ *is a cyclic group. What are its generators?* CO4 K2

15*(a) Define Eulerian and Hamiltonian graphs with example.* CO5 K2

OR

(b) Show that the number of vertices of odd degree in an undirected graph is even. CO5 K2

SECTION – C

Answer either (a) or (b) in each Question **(4 x 10=40)**

16*(a) Find whether the conclusion C:* $\neg P$ *follows from the premises* $H_1: \neg(P \wedge \neg Q)$; $H_2: \neg R$; $H_3: Q \rightarrow R$ *using truth table technique.* CO1 K3

OR

(b) Obtain $the\ PDNF\ and\ PCNF\ of\ (\neg P \rightarrow \neg R) \wedge (P \leftrightarrow Q)$ CO1 K3

17*(a) If R is the relation on the set of ordered pairs of positive integers such that* $(a,b)\ R\ (c,d)$ *if and only if* $ad = bc$ *Show that R is an equivalence relation.* CO2 K3

OR

(b) Prove that $(A \oplus B) \times C = (A \times C) \oplus (B \times C)$ CO2 K3

18*(a) Prove that* D_{42} *is a complemented lattice by finding the complements of all the elements.* CO3 K3

OR

(b)Simplify the Boolean function $f(a,b,c,d) = \sum(0,1,2,3,4,5,6,7,8,9,11)$ CO3 K3

19*(a) show that the set* Q^+ *of all positive rational numbers forms an abelian group under the operation * defined by* $a * b = \frac{ab}{2}; a, b \in Q^+$ CO4 K3

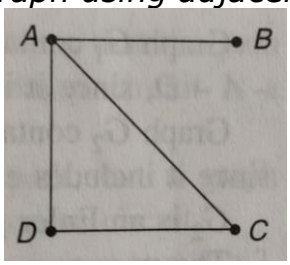
OR

(b) List the code words generated by the encoding function $e: B^3 \rightarrow B^6$ *whose parity check matrix is* $H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ CO4 K3

SECTION – D

Answer either (a) or (b) **(1 x 10=10)**

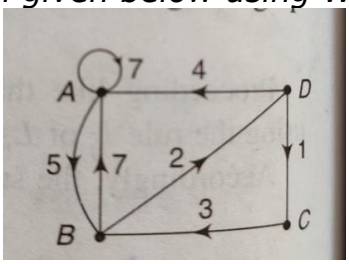
20*(a) summarize all the paths of length 4 between B and D for the following graph using adjacency matrix*



CO5 K4

(OR)

(b) Evaluate the shortest distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the weighted graph given below using Warshall's algorithm.



CO5 K4
